

Samaranjan Manjhi

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EDUCATION

Degree	University	Institute	Year	CPI/%
Graduation	Mumbai University	Shree LR Tiwari College Of Engineering, Thane	2019-23	8.56
Intermediate/+2	HSC	Thakur College Of Science & Commerce, Mumbai	2017-19	74.00
Matriculation	SSC	Thakur Shyamnarayan High School, Mumbai	2016-17	89.60

PROFESSIONAL EXPERIENCE

Programmer

August 2024 – Present

MicroWorld Software Services Pvt Ltd

Andheri, Mumbai

- Developed system utilities & security modules (vulnerability checker, FIM, Process Monitor, content scanner, ransomware defense) using C++/Linux/macOS APIs.
- Integrated security libraries & kernel-level monitoring (OpenSSL, cURL, Boost, Tesseract, Poppler) to enhance EDR solutions.

Trainee Programmer

August 2023 – August 2024

MicroWorld Software Services Pvt Ltd

Andheri, Mumbai

- Built cross-platform Qt GUI tools in C++ for Linux and macOS.
- Gained hands-on experience with Git, Xcode, and build systems while assisting in file scanning utilities.

KEY PROJECTS

ProcMon - Process Monitoring Tool | C/C++, libmagic, bsm

July 2025 - Sept 2025

- Developed a lightweight macOS process monitoring tool that logs new executions (e.g., scp, rsync, vim, file copy) with command-line arguments and process hierarchies in real time using system calls, without relying on Endpoint Security or external libraries.

File Integrity Monitoring System | C/C++, OpenSSL, sqlite3

June 2025 - Aug 2025

- Developed a macOS File Integrity Monitoring (FIM) tool that loads directory watch lists from config, stores metadata (permissions, MD5, size, timestamps) in SQLite, and continuously monitors for real-time changes (permission/content/deletion) with alerting and event logging for security auditing.

cBackup - Cloud Backup & Monitoring Tool | C/C++, OpenSSL, cURL

Feb 2025 - May 2025

- Developed cBackup (Desktop EDR Edition), a cross-platform backup and monitoring tool for macOS/Linux that tracks file transfers (USB, scp, rsync, curl, email), auto-backs up files locally (.attach/_usb), and synchronizes them across multiple cloud services with 10-minute cycles ensuring consistency and data loss prevention.

Content Scanner | C/C++, Leptonica, Tesseract, Poppler

Sept 2024 - Dec 2024

- Developed a data scanning tool that identifies confidential data (PAN, emails, etc.) across 12+ file types (PDF, images, XML, DOCX, archives) using regex patterns; deployable as both a shared library and a CLI for versatile integration.

System Call Interception Module | C, Linux Kernel, Hooking

Mar 2024 - Aug 2024

- Developed a Linux kernel module to intercept system calls and implement custom hooks for monitoring file operations by extension (creation, renaming, moving), strengthening file system protection and enhancing security against ransomware and malicious activities.

Vulnerability Checker Application | C/C++, Qt, Boost, cURL, OpenSSL, cJSON

Oct 2023 - Jan 2024

- Developed a secure vulnerability checker application by integrating front-end and back-end technologies, leveraging cURL for ZIP downloads, XML parsing, and efficient SQLite storage to ensure seamless functionality, organized scanning, and reliable data management.

TECHNICAL SKILLS

Languages: C/C++

Frameworks: Qt

Operating System: Linux, macOS

Developer Tools: Vim, Xcode, Git, SQLite

Libraries: boost, cJSON, curl, leptonica, tesseract, poppler, libmagic, libarchive